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IODINE-125 EPISCLERAL EYE PLAQUE RADIOTHERAPY IN TREATMENT OF CIRCUMPAPILLARY CHOROIDAL MELANOMA

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Ninety patients with the diagnosis of choroidal melanoma have been treated with I-125 episcleral eye plaque radiotherapy at this institution since 1985. Of these ninety patients, thirty-five also underwent adjuvant hyperthermia with eye plaque placement. Approximately thirty percent of the tumors were circumpapillary and/or juxtapapillary in nature. This poster will present the treatment technique of the circumpapillary tumors in comparison to more peripheral choroidal melanomas. Results in effectiveness of therapy are presented on the basis of visual acuity changes, topographical changes in the tumor itself, local recurrence, distant metastases and early survival data. Complications of therapy are also compared in the two groups.

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EPISCLERAL PLAQUE RADIOTHERAPY USING RADIOACTIVE 198-AU IN THE TREATMENT OF CHOROIDAL MELANOMAS

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Purpose/Objective: This is a retrospective review of patients with choroidal melanoma treated with episcleral plaque radiotherapy using radioactive 198-Au in order to analyze the results and complications of therapy.

Materials & Methods: Between 1973 and 1991, 123 patients with choroidal melanoma were treated with episcleral plaque radiotherapy using radioactive 198-Au at Baylor College of Medicine by one ophthalmologist. One hundred patients were available for evaluation. Mean follow-up time was 59 months (range 6-225 months). The T-stage was: T1 (3%), T2 (43%), T3 (54%). The mean tumor elevation was 5.89 mm (range 2-12 mm). The dose was prescribed to the tumor apex with the mean dose being 160.2 Gy (range 102 Gy-200 Gy).

Results: A decrease in tumor elevation was seen in 82% of the patients. Five-year local control rate was 83%. Twenty percent of the patients had local recurrence requiring enucleation. Distant metastasis was seen in 7% of the patients. Actuarial disease-free survival, including salvage, is 92% at five years. Useful vision (> 5/200) was maintained in 62% of the patients. Vision was significantly influenced by T-stage, T3 worse than T2 (p = 0.05). Fifty-seven percent of the patients had treatment complications including cataracts, subretinal hemorrhage, sclera ulceration, and glaucoma. Six patients underwent enucleation because of radiotherapy complications but melanoma was found pathologically in the eye of all six cases. Risk of complications and enucleation was significantly affected by T-stage (T3 worse than T2, p = .0016, p = 0.036 respectively). Dose (within range mentioned), age, sex, and location did not affect tumor local control, risk of complication or enucleation.

Conclusion: (1) Episcleral plaque radiotherapy using 198-Au is an effective therapeutic alternative to enucleation. (2) Although complications were encountered, useful vision was maintained in two-thirds of the patients. (3) Tumor control rate and risk of complications were significantly affected by T-stage.